Bioenergy

pyrolysis, gasification, and anaerobic digestion

Rafael Nogueira Nakashima

Department of Energy Conversion and Storage
Technical University of Denmark

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Outline

Biomass

Thermochemical conversion processes
- Combustion
- Pyrolysis
- Gasification

Break

Other conversion processes
- Biological process example: anaerobic digestion
- Hybrid process example: bio-e-fuels

Activities

Feedback

Learning objectives

By the end of this lecture, you should be able to:

- Identify true or false statements related to biomass energy, conversion of biomass, and bio-e-fuels.
- Calculate the heating value of biomass resources and biomass conversion reactions.

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What is biomass?

more than just wood

Definition

Biomass is any organic matter derived from living beings, excluding fossil fuels.



Figure: Wood chips [1]

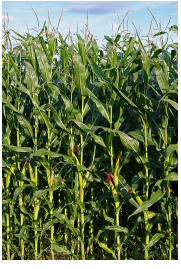


Figure: A maize plantation [2]



Figure: Manure [3]

Photosynthesis

how plants store energy from sunlight

$$6\,H_2O_{(g)} + 6\,CO_{2(g)} + \text{Energy} \underset{\text{respiration}}{\overset{\text{photosynthesis}}{\rightleftharpoons}} C_6H_{12}O_{6(s)} + 6\,O_{2(g)}$$

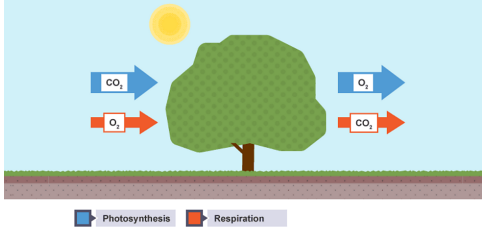


Figure: Plants during the day [4]

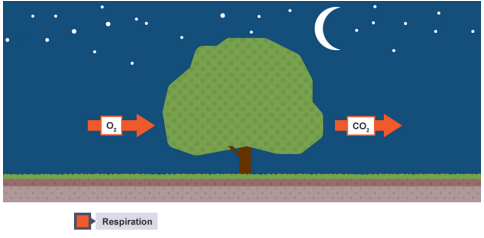


Figure: Plants during the night [4]

Biomass conversion efficiency

from sunlight to organic molecules

The energy efficiency conversion from sunlight to biomass is relatively low ($< 5\%$), compared with photovoltaics (20-25%);

Inefficiency sources

- Capturing sunlight
- Absorbing sunlight
- Converting sunlight
- Respiration

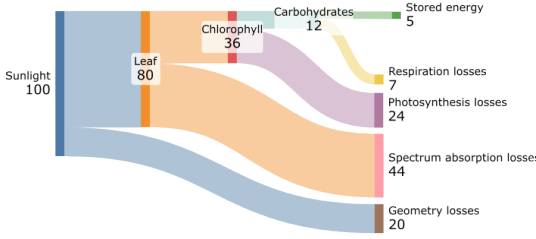


Figure: Energy storage efficiency for an ideal plant [5]

Advantage of biomass

dual function: carbon capture + energy storage

Main advantages

- Energy storage: energy is already stored in biomass;
- Carbon capture: biomass holds carbon that can be directly converted to fuels;

Conversion routes

- Thermochemical: combustion, pyrolysis, gasification.
- Biological: fermentation, anaerobic digestion, composting

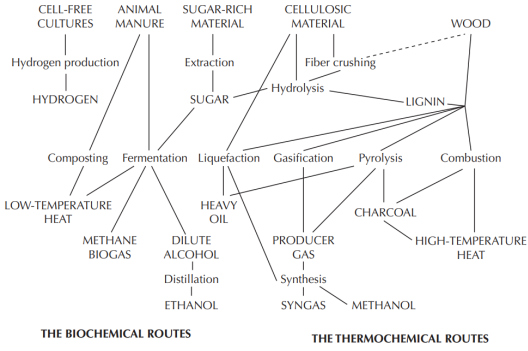


Figure: Different uses of biomass [5]

Activity

Biomass in Denmark

Problem

What is the main crop planted in the agricultural areas of Denmark and for which purpose it mainly is?

a. *Rye for human consumption (i.e., rye bread)*
b. *Cereals for pigs, cattle, and chickens.*
c. *Wood for furniture, lumber and energy.*
d. *Potatoes for human consumption and export.*
e. *Sugar beets for sugar production and export.*

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Recap: Heating value

energy released from combustion

Combustion of hydrogen example[1]:

$$\underbrace{H_{2(g)}\big|_{25^\circ C,1bar}}_{h=0} + \frac{1}{2}\underbrace{O_{2(g)}\big|_{25^\circ C,1bar}}_{h=0} \rightarrow \underbrace{H_2O_{(g)}\big|_{25^\circ C,1bar}}_{h=-241830\text{J/mol}} + \dot{Q}_{lhv} \rightarrow \underbrace{H_2O_{(l)}\big|_{25^\circ C,1bar}}_{h=-285830\text{J/mol}} + \dot{Q}_{hhv}$$

Lower heating value is the reaction heat assuming water vapour as the final product:

$$LHV = \dot{Q}_{lhv} = \cancel{h_{H_{2(g)}}}^{0} + \cancel{h_{O_{2(g)}}}^{0} - h_{H_2O_{(g)}} = 241.83\ kJ/mol$$

Higher heating value is the reaction heat assuming liquid water as the final product:

$$HHV = \dot{Q}_{hhv} = \dot{Q}_{lhv} + (h_{H_2O_{(g)}} - h_{H_2O_{(l)}}) = 285.83\ kJ/mol$$

[1] $H_2O(g)$ @ 25 °C and 1 bar is a hypothetical ideal gas state (not real)

Activity

Coal vs. biomass

Problem

What is the released heat (LHV basis) of fructose ($C_6H_{12}O_6$) in kJ/mol?

$$LHV_{C_6H_{12}O_6} = ?$$

a *2,810,440*

b *2,546,440*

c *2,810.44*

d *2,546.44*

e *I don't know*

Molecule	$h_{25^\circ C,1bar}$
C	0
$C_6H_{12}O_6$	-1265.6 kJ/mol
CO_2	-393.51 kJ/mol
H_2	0
H_2O_g	-241.83 kJ/mol
H_2O_l	-285.83 kJ/mol
O_2	0

Table: Enthalpies for selected molecules

A biomass-based power plant

heating a steam cycle that converts heat to power

- Biomass is treated to improve combustion (e.g., drying or shape)
- Biomass combustion heats up high-pressure steam
- Steam can be used to generate power and/or heat.
- Combustion gases are treated to remove pollutants
- Biomass is reduced to ashes.

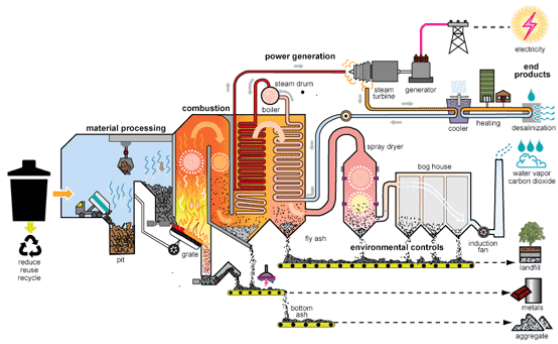


Figure: A waste-to-energy plant (very similar to a biomass-based one) [8]

Activity

Endoreversible processes

Problem

The equivalent of 1 MW of biomass (LHV basis) is burned in a boiler at 1500 K. The released heat is used to heat up steam up to 300 °C. The steam is used to generate power and later cooled down to a minimal temperature of 40 °C, by exchanging heat with the environment at 25 °C. Assuming that the steam cycle is reversible, what is the power generated by it?

a *0.98 MW*

b *0.92 MW*

c *0.80 MW*

d *0.45 MW*

e *0.35 MW*

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What happens if biomass is heated?

avoiding combustion drawbacks

Biomass can be converted into char, bio-oil or synthesis gas (syngas) by being exposed to heat.

Reaction examples

- Char: $C_6H_{12}O_6 + \text{heat} \rightarrow \underbrace{6C}_{char} + 6H_2O$
- Bio-oil: $C_6H_{12}O_6 + \text{heat} \rightarrow \underbrace{6CH_2O}_{bio-oil}$
- Syngas: $C_6H_{12}O_6 + \text{heat} \rightarrow \underbrace{6CO + 6H_2}_{syngas}$



Figure: A pizza left in the oven for too long has suffered a pyrolysis process [9]

Pyrolysis

an endothermic process (needs heat)

The main product is char (i.e., biochar). Heat energy usually derives from burning the syngas. Bio-oil is an attractive by-product to blend with fossil fuels.

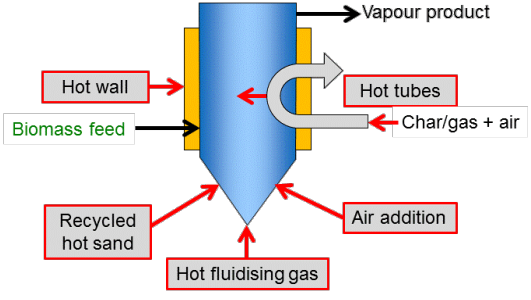


Figure: Heat delivery methods in pyrolysis [11]

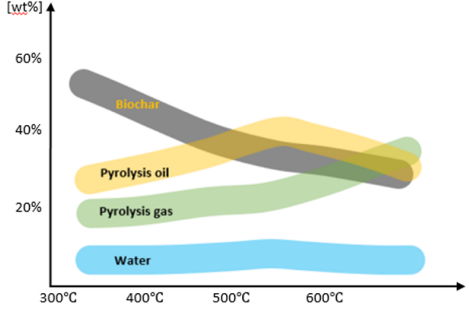


Figure: Temperature effect in pyrolysis [12]

Bio-char

a cheap alternative to carbon capture?

Bio-char can be sold to farmers for soil improvement and to business as carbon credits.

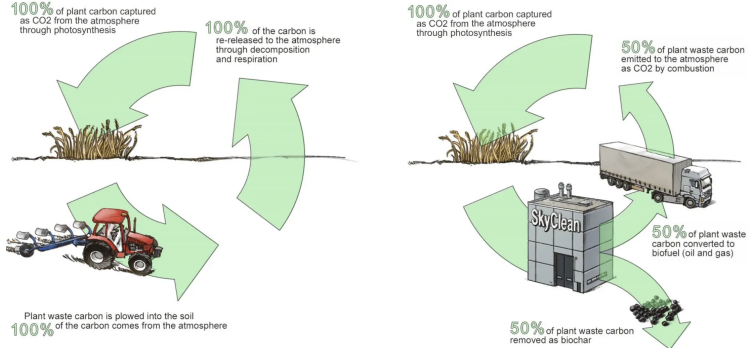


Figure: Stiesdal business model for pyrolysis [13]

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Gasification

pushing biomass conversion to gas production

Air, oxygen, and/or water are added to the reactor to convert char into syngas, making the reaction exothermal.

$$\lambda = \frac{\dot{m}_{air,actual}}{\dot{m}_{air,st.}} \quad (1)$$

- λ: air-fuel equivalence ratio
- $\dot{m}_{air,actual}$: actual air mass flow rate [kg/s]
- $\dot{m}_{air,st.}$: stoichiometric air mass flow rate [kg/s]

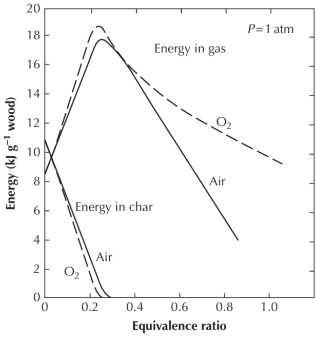


Figure: Energy distribution vs. equivalence ratio [5]

Why syngas?

producing chemicals and efficient power

Synthesis gas ($CO+H_2$) can be directly converted into power with gas turbines ($\uparrow \eta$) or chemicals (e.g., fuels or plastics).

Reaction examples: Syngas to chemicals

$$CO + 3H_2 \rightarrow \underbrace{CH_4}_{\text{natural gas}} + H_2O$$

$$CO + 2H_2 \rightarrow \underbrace{CH_3OH}_{\text{methanol}} + H_2O$$

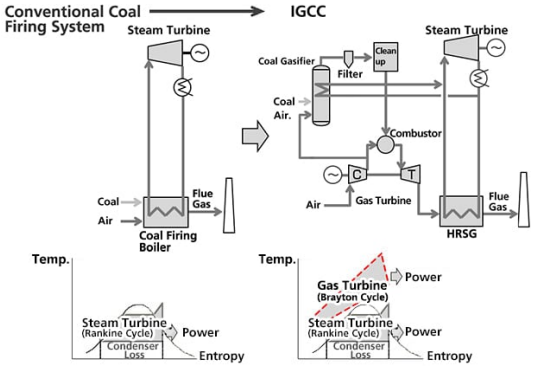


Figure: Gasification raises the high temperature of the heat cycle [15]

Activity

How much fuel can we produce?

Problem

Considering the following partial oxidation reaction from a gasifier:

$$C_6H_{12}O_6 + 2\,O_2 \longrightarrow 4\,CO + 4\,H_2 + 2\,CO_2 + 2\,H_2O$$

If the syngas is used to produce methanol according to the following reaction:

$$CO + 2\,H_2 \longrightarrow CH_3OH$$

How much energy can be converted from biomass to methanol in LHV basis? Consider $LHV_{C_6H_{12}O_6} = 2546.2$ kJ/mol, $LHV_{CH_3OH} = 638.1$ kJ/mol, and exclude other reactions.

a 100 %
b 75 %
c 50 %
d 25 %
e 0 %

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What if we have too much water?

biological conversion of organic wastes

In some cases, biomass is available with a high concentration of water (e.g., animal waste and industrial wastewater).

- Removing water for thermochemical conversion is expensive
- Biological conversion can reduce environmental danger and produce renewable fuel.



Figure: A large lagoon containing animal wastes [10]

Anaerobic digestion

bacteria converting biomass without oxygen

A simplified theoretical reaction (Buswell equation):

$$\underbrace{C_c H_h O_o}_{biomass} + \left(c - \frac{h}{4} - \frac{o}{2}\right) H_2O \rightarrow \underbrace{\left(\frac{c}{2} + \frac{h}{8} - \frac{o}{4}\right) CH_4 + \left(\frac{c}{2} - \frac{h}{8} + \frac{o}{4}\right) CO_2}_{biogas}$$

Example: Fructose anaerobic digestion

$$\underbrace{C_6H_{12}O_6}_{fructose} \rightarrow \underbrace{3CH_4 + 3CO_2}_{biogas}$$

Important

You NEED to use the Buswell equation for estimating the biogas production. You may get a WRONG answer if you don't use it.

Biogas applications

Combined heat and power generation and natural gas production

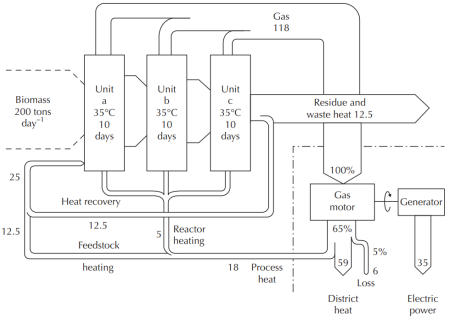


Figure: Power generation from biogas [14]

1. Reactors convert biomass into gas in a slow biological process
2. A significant portion of biomass cannot be converted into gas (i.e., complex carbohydrates);
3. Gas is converted into power and heat using a gas engine
4. Part of the heat is used to keep reactors warm

Biogas applications

Combined heat and power generation and natural gas production

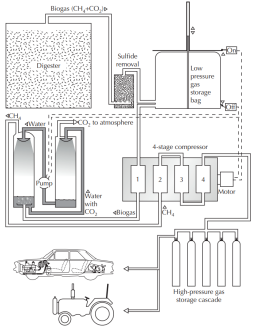


Figure: Biomethane production from biogas [14]

Alternatively, biogas can be purified to be sold as a natural gas substitute:

- 3b CO_2 is removed from biogas in a separation step (e.g., pressurised water absorption)
- 4b Purified methane is compressed and can be used/sold as a natural gas substitute.

Photos

Biogas plant, power generation and separation units



Figure: Biogas plant



Figure: Biogas engine



Figure: Biogas purification unit (PSA)

Activity

Biogas efficiency

Problem (2025 exam edited)

An anaerobic digester converts wastewater containing fructose into biogas following the Buswell equation. If fructose ($C_6H_{12}O_6$) is the input and methane (CH_4) is the output, what is the energy efficiency of this process on a lower heating value (LHV) basis? Assume that the LHV of fructose is 2546 kJ/mol and the LHV of methane is 802 kJ/mol. Assume complete conversion and neglect energy losses other than chemical conversion

a 100 %
b 95 %
c 63 %
d 32 %
e 0 %

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Integration using hydrogen

boosting biofuel production

In our previous example, we could not convert all carbon into fuel (CH_4):

$$C_6H_{12}O_6 \rightarrow 3CH_4 + 3CO_2$$

But if we add up hydrogen, we can convert the remaining CO_2 into methane:

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$$

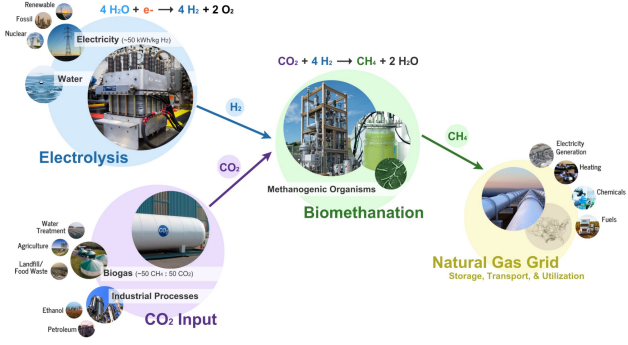


Figure: Biomethanation concept [16]

Other examples

different carbon-based fuels

From leftover carbon monoxide and/or carbon dioxide of syngas/biogas, we can produce different products if added hydrogen from electrolysis.

(+) More product per biomass (less land use)

(-) More energy demand (higher cost of final product)

Methane

$$CO + 3\,H_2 \longrightarrow CH_4 + H_2O$$
$$CO_2 + 4\,H_2 \longrightarrow CH_4 + 2\,H_2O$$

Methanol

$$CO + 2\,H_2 \longrightarrow CH_3OH$$
$$CO_2 + 3\,H_2 \longrightarrow CH_3OH + H_2O$$

DME - Dimethyl ether

$$3\,CO + 3\,H_2 \longrightarrow CH_3OCH_3 + CO_2$$
$$2\,CO_2 + 6\,H_2 \longrightarrow CH_3OCH_3 + 3\,H_2O$$

Activity

Bio-e-fuels

Problem

Considering the last example for a gasification process:

$$C_6H_{12}O_6 + 2\,O_2 \longrightarrow 4\,CO + 4\,H_2 + 2\,CO_2 + 2\,H_2O$$

How much more methanol can we produce, compared to the last activity, if we add extra hydrogen from electrolysis to complete the reactions?

$$CO + 2\,H_2 \longrightarrow CH_3OH$$
$$CO_2 + 3\,H_2 \longrightarrow CH_3OH$$

a 400 % more
b 300 % more
c 200 % more
d 100 % more
e 50 % more

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Let's vote!

Problem

Should we subsidise the production of biogas to achieve a 100% green gas grid by 2030, even if this relies on expanding animal farming (i.e., cows and pigs)?

a *Yes*

b *No*

Note

You will have to defend your position afterwards.

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Muddy points and key insights

Feedback for your learning

Here is the chance for you to get feedback on your learning.

The quiz will be open only TODAY.

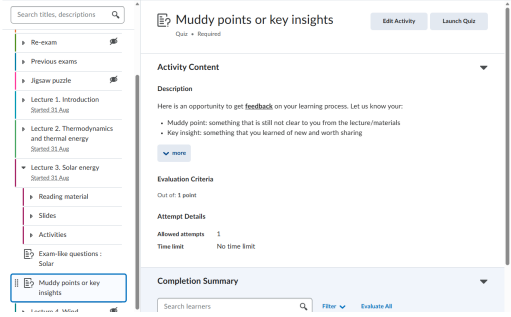


Figure: A screenshot capture of your DTU Learn

References I

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